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Tractor

Abstract

A tractor includes: a vehicle body; an engine compartment; a driving unit that: is disposed behind the engine compartment, and includes: a brake pedal that includes: a support shaft; and a pedal arm portion; a master cylinder that is operated by the brake pedal and disposed further toward a vehicle-body back side than the support shaft; and a return spring that biases the brake pedal to swing back to a brake off position; and a partition that: is disposed further toward a vehicle-body front side than the brake pedal, and separates the engine compartment and the driving unit from each other, the return spring being engaged to the partition and to the pedal arm portion.

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Background/Summary

BACKGROUND
Technical Field
(1) The present invention relates to a tractor.
Description of Related Art
(2) As illustrated in patent literature 1 for example, among tractors, there are tractors that are provided with a driving unit having a brake pedal, a master cylinder operated by the brake pedal, and a return spring that biases the brake pedal to swing back to a brake off position. In these tractors, the master cylinder is provided on a vehicle-body back side of a support shaft (horizontal shaft) of the brake pedal, and the return spring is engaged to a pedal arm portion (swing arm) of the brake pedal and to a cylinder case.
PRIOR-ART LITERATURE
(3) [Patent Literature 1] JP S57-9656 U
(4) Conventionally, the return spring is engaged to the pedal arm portion (swing arm) of the brake pedal and to the cylinder case. As such, the return spring enters a state of being extended to a master-cylinder side from the pedal arm portion of the brake pedal.
SUMMARY
(5) One or more embodiments of the present invention provide a tractor that can keep a return spring from extending to a master-cylinder side while providing the master cylinder on a vehicle-body back side of a support shaft of a brake pedal and keep a structure of engaging the return spring simple.
(6) A tractor of one or more embodiments of the present invention is provided with: a vehicle body; an engine compartment; and a driving unit that is provided behind the engine compartment and has a brake pedal, a master cylinder operated by the brake pedal, and a return spring that biases the brake pedal to swing back to a brake off position; wherein the master cylinder is disposed further toward a vehicle-body back side than a support shaft of the brake pedal, provided is a partition member (i.e., a partition) that is disposed further toward a vehicle-body front side than the brake pedal and separates the engine compartment and the driving unit from each other, and the return spring is engaged to the partition member and to a pedal arm portion of the brake pedal.
(7) According to the present configuration, the return spring engages to the pedal arm portion and the partition member disposed further toward the vehicle-body front side than the brake pedal. As such, the return spring is kept from extending to a master-cylinder side from the pedal arm portion. The partition member is used as a member whereto the return spring is engaged. As such, a structure of engaging the return spring is obtained in a simple state requiring no special engaging

member.
(8) In one or more embodiments of the present invention, a portion of the partition member corresponding to a footboard of the brake pedal is disposed further toward a vehicle-body front side than a portion of the partition member corresponding to the support shaft.
(9) According to the present configuration, a space wherein the brake pedal is depressed and moved can be extended to a vehicle-body front side further than the support shaft. As such, a footwell of the driving unit can be widened.
(10) In one or more embodiments of the present invention, the return spring is constituted by a torsion coil spring, and the torsion coil spring is supported by the support shaft with a coil portion being engaged with the support shaft.
(11) According to the present configuration, the coil portion is supported by the support shaft, and the return spring stably imparts to the brake pedal a biasing force of biasing the brake pedal to the brake off position. As such, the brake pedal can be accurately biased to the brake off position by the return spring without requiring a special support member supporting the coil portion.
(12) In one or more embodiments of the present invention, a first arm extending from one end side of the torsion coil spring is engaged with the partition member and a second arm extending from another end side of the torsion coil spring is engaged with the pedal arm portion of the brake pedal, the second arm on the other end side is provided with a bent end portion that engages with the pedal arm portion, the pedal arm portion is provided with a through hole which the bent end portion engages, and the through hole is configured such that the bent end portion can be inserted into the through hole in a direction along a shaft core of the support shaft.
(13) According to the present configuration, the torsion coil spring biases the brake pedal to swing by using the partition member as a reaction-force point. As such, the brake pedal can be accurately biased to the brake off position by the torsion coil spring.
(14) According to the present configuration, an assembly method can be adopted wherein as the brake pedal is fitted to the support shaft, the bent end portion enters the through hole and engages the pedal arm portion. This facilitates assembly of the brake pedal.

Description

BRIEF DESCRIPTION OF DRAWINGS
(1) FIG. **1** is a left side view illustrating an entirety of a tractor.
(2) FIG. **2** is a left side view illustrating a first exhaust-gas cleaning device and a second exhaust-gas cleaning device.
(3) FIG. **3** is a plan view illustrating the first exhaust-gas cleaning device and the second exhaust-gas cleaning device.
(4) FIG. **4** is a back view illustrating a support member.
(5) FIG. **5** is a perspective view of the support member in an exploded state.
(6) FIG. **6** is a right side view illustrating left and right brake operation devices.
(7) FIG. **7** is an explanatory diagram illustrating a summary of assembly of a brake pedal.
(8) FIG. **8** is a back view illustrating a steering-post cover and a panel cover.
(9) FIG. **9** is a back view illustrating a panel cover provided with another example of one or more embodiments.
(10) FIG. **10** is a perspective view of a steering-post cover provided with another example of one or more embodiments.
DETAILED DESCRIPTION OF EMBODIMENTS
(11) Embodiments of the present invention are described below based on the drawings.
(12) Note that in the following description, in regards to a traveling vehicle body of a tractor, the direction of arrow F illustrated in FIG. **1** is defined as “vehicle-body front”, the direction of arrow

B is defined as "vehicle-body back", the direction of arrow U is defined as "vehicle-body up", the direction of arrow D is defined as "vehicle-body down", the direction heading toward the surface of the page is defined as "vehicle-body left", and the direction heading toward the reverse face of the page is defined as "vehicle-body right".
(13) [Overall Configuration of Tractor]
(14) As illustrated in FIG. **1**, the tractor is provided with a traveling vehicle body **3** supported by a pair of left and right front wheels **1**, which can be steered and driven, and a pair of left and right back wheels **2**, which can be driven. A vehicle body frame **4** of the traveling vehicle body **3** is constituted by an engine **5**, a flywheel housing **6** connected to a back portion of the engine **5**, a clutch housing **7** connected to a back portion of the flywheel housing **6**, a transmission case **7***a* connected to a back portion of the clutch housing **7**, and a front frame **8** connected to a lower portion of the engine **5**. A motor unit **9** provided with the engine **5** is formed in a front portion of the traveling vehicle body **3**. A driving unit **12**, which is provided with a driver's seat **10** and with a steering wheel **11** whereby a steering operation of the front wheels **1** is performed, is formed in a back portion of the traveling vehicle body **3**. A linking mechanism **13**, which connects a work apparatus such as a rotary tilling apparatus (not illustrated) in a manner enabling a raising and lowering operation of the work apparatus, and a power takeoff shaft **14**, which takes power from the engine **5** and outputs this to the connected work apparatus, are provided in a back portion of the transmission case **7***a*. Reference numeral **20** illustrated in FIG. **1** is a ROPS frame.
(15) [Configuration of Motor Unit]
(16) As illustrated in FIGS. **1** and **2**, the motor unit **9** is provided with an engine compartment **15**. The engine compartment **15** is formed by an engine bonnet **16**, which covers the engine compartment **15** from above and the front; a partition member **17** that forms a partition between the engine compartment **15** and the driving unit **12**; and the like.
(17) As illustrated in FIGS. **1** and **2**, the engine **5**, a radiator **18** that cools the engine **5**, and a first exhaust-gas cleaning device (DPF) **21** and second exhaust-gas cleaning device (SCR) **22** that perform cleaning processes of exhaust gas exhausted by the engine **5** are provided in the engine compartment **15**.
(18) [Configuration of Radiator]
(19) As illustrated in FIG. **1**, the radiator **18** is provided in front of the engine **5**. A blowing action of a rotary fan **19** positioned between the radiator **18** and the engine **5** introduces cooling air from outside the engine compartment **15** to inside the engine compartment and supplies the cooling air to the radiator **18**. The cooling air is supplied to the radiator **18** in a state of the cooling air passing through the radiator **18** from front to back. In the radiator **18**, engine cooling water is cooled by heat exchange between the supplied cooling air and the engine cooling water. The engine **5** is cooled by the cooled engine cooling water being supplied to the engine **5**.
(20) [Configurations of First Exhaust-Gas Cleaning Device and Second Exhaust-Gas Cleaning Device]
(21) The engine **5** is a diesel engine. As illustrated in FIGS. **2** and **3**, the first exhaust-gas cleaning device **21** is provided above the engine **5** in a state wherein a long side of the first exhaust-gas cleaning device is substantially parallel to a vehicle-body horizontal direction (or a vehicle-body lateral width direction). A maximum length H in a vehicle-body vertical direction (or a vehicle-body up-down direction) of the first exhaust-gas cleaning device **21** is set to be shorter than a maximum length W in a vehicle-body longitudinal direction (or a vehicle-body front-back direction) of the first exhaust-gas cleaning device **21**. As illustrated in FIGS. **2** and **3**, an exhaust-gas suctioning portion **21***a*, which is provided to a part on one end side, in a vehicle-body horizontal width direction, of the first exhaust-gas cleaning device **21**, and an exhaust-gas exhausting portion **5***a*, which is provided to the engine **5**, are connected. An exhaust-gas discharging portion **21***b* is provided to a part on another end side, in the vehicle-body horizontal width direction, of the first exhaust-gas cleaning device **21**. In one or more embodiments, the

exhaust-gas suctioning portion **21***a* protrudes downward from a lower portion of the first exhaust-gas cleaning device **21** at an end portion on a vehicle-body left horizontal side of the first exhaust-gas cleaning device **21**, and the exhaust-gas discharging portion **21***b* protrudes backward from an end portion on a vehicle-body right horizontal side of the first exhaust-gas cleaning device **21**.
(22) In the first exhaust-gas cleaning device **21**, the exhaust gas exhausted by the engine **5** from the exhaust-gas exhausting portion **5***a* is suctioned by the exhaust-gas suctioning portion **21***a* into the apparatus, and diesel microparticles included in the suctioned exhaust gas are collected by a collection filter (not illustrated). This performs an exhaust-gas cleaning process of decreasing the diesel microparticles. The exhaust gas subjected to the cleaning process is discharged from the exhaust-gas discharging portion **21***b*.
(23) As illustrated in FIGS. **2** and **3**, the second exhaust-gas cleaning device **22** is provided behind the engine **5** in a state wherein a long side of the second exhaust-gas cleaning device is substantially parallel to the vehicle-body horizontal direction. An exhaust-gas introduction portion **22***a*, which is provided to a part on one end side, in the vehicle-body horizontal width direction, of the second exhaust-gas cleaning device **22**, and the exhaust-gas discharging portion **21***b* of the first exhaust-gas cleaning device **21** are connected by a connecting pipe **23**. An exhaust-gas discharging portion **22***b* is provided to a part on another end side, in the vehicle-body horizontal width direction, of the second exhaust-gas cleaning device **22**. In one or more embodiments, the exhaust-gas introduction portion **22***a* is provided to an end portion on a vehicle-body left horizontal side of the second exhaust-gas cleaning device **22**, and the exhaust-gas discharging portion **22***b* is provided to an end portion on a vehicle-body right horizontal side of the second exhaust-gas cleaning device **22**.
(24) In the second exhaust-gas cleaning device **22**, the exhaust gas discharged by the first exhaust-gas cleaning device **21** from the exhaust-gas discharging portion **21***b* is supplied by the connecting pipe **23** to the exhaust-gas introduction portion **22***a* and introduced by the exhaust-gas introduction portion **22***a* into the apparatus, and the introduced exhaust gas is subjected to the cleaning process by a reducing agent. Specifically, aqueous urea as the reducing agent is injected into the introduced exhaust gas, hydrolyzing the exhaust gas. This performs an exhaust-gas cleaning process of decreasing nitrogen oxides included in the exhaust gas. The exhaust gas subjected to the cleaning process is exhausted from the exhaust-gas discharging portion **22***b* to a vehicle-body horizontal outer side.
(25) As illustrated in FIG. **3**, the first exhaust-gas cleaning device **21** is supported in an attachment attitude wherein the part on the other end side, in the vehicle-body horizontal width direction, is swung and displaced toward a vehicle-body front side with respect to the part on the one end side, in the vehicle-body horizontal width direction, in a state wherein the exhaust-gas suctioning portion **21***a* is a swinging shaft. That is, the first exhaust-gas cleaning device **21** is supported in an attitude wherein in a plan view, the longitudinal direction thereof is inclined relative to the vehicle-body horizontal width direction. As illustrated in FIG. **2**, the exhaust-gas introduction portion **22***a* of the second exhaust-gas cleaning device **22** extends diagonally backward and upward from the second exhaust-gas cleaning device **22**. A positional relationship between the exhaust-gas discharging portion **21***b* of the first exhaust-gas cleaning device **21** and the exhaust-gas introduction portion **22***a* of the second exhaust-gas cleaning device **22** can be made to be a positional relationship appropriate for adopting a connecting pipe **23** having no bellows or other adjustment means of adjusting the positional relationship between the exhaust-gas discharging portion **21***b* and the exhaust-gas introduction portion **22***a*. This can be done by bringing the first exhaust-gas cleaning device **21** and the second exhaust-gas cleaning device **22** in proximity to each other in the vehicle-body longitudinal direction and by directly connecting the exhaust-gas suctioning portion **21***a* of the first exhaust-gas cleaning device **21** to the exhaust-gas exhausting portion **5***a* of the engine **5**.
(26) As illustrated in FIGS. **2** and **3**, a back portion **21***c* of the first exhaust-gas cleaning devices **21** is positioned on a vehicle-body back side of a back end portion of the engine **5**. In a plan view, the

back portion **21***c* of the first exhaust-gas cleaning device **21** and the second exhaust-gas cleaning device **22** overlap.
(27) As illustrated in FIGS. **2** and **3**, the partition member **17** that forms the partition between the engine compartment **15** and the driving unit **12** is provided across a back location of the second exhaust-gas cleaning device **22** and a lower location of the second exhaust-gas cleaning devices **22**. The partition member **17** is configured so a part **17***a*— positioned in the lower location of the second exhaust-gas cleaning device **22**—of the partition member **17** is positioned further toward a vehicle-body front side than a part **17***b*—positioned in the back location of the second exhaust-gas cleaning device **22**—of the partition member **17**. The part **17***a*—positioned in the lower location of the second exhaust-gas cleaning device **22**—of the partition member **17** is provided with a portion in an inclined state that, in moving toward its lower end side, is positioned more to a vehicle-body front side. A footwell of the driving unit **12** can be extended below the second exhaust-gas cleaning device **22**.
(28) As illustrated in FIG. **6**, electrical wires **24** are provided across the engine compartment **15** and the driving unit **12**. In one or more embodiments, a plurality of electrical wires **24** is provided. However, it is also possible for only one to be provided. The electrical wires **24** enter from the engine compartment **15** into the footwell of the driving unit **12** by passing below a part positioned in a central portion in the vehicle-body horizontal width direction of the partition member **17**. The electrical wires **24** transmit information relating to actuation of the engine **5** to gauges of the driving unit **12**. A portion of the electrical wires **24** that is positioned on a driving-unit side of the partition member **17** is wired in a state of conforming to the part **17***a*—positioned in the lower location of the second exhaust-gas cleaning device **22**—of the partition member **17** in the footwell of the driving unit **12**. The electrical wires **24** pass through a conduit **25**. The conduit **25** is supported on the partition member **17** by a clamp **26**.
(29) [Support Member of Exhaust-Gas Cleaning Devices]
(30) As illustrated in FIGS. **2**, **3**, and **4**, the first exhaust-gas cleaning device **21** and the second exhaust-gas cleaning device **22** are supported on the vehicle body frame **4** via a support member **30**. The first exhaust-gas cleaning device **21** and the second exhaust-gas cleaning device **22** are supported by the vehicle body frame **4** in a state wherein the positional relationship between the exhaust-gas discharging portion **21***b* of the first exhaust-gas cleaning device **21** and the exhaust-gas introduction portion **22***a* of the second exhaust-gas cleaning device **22** is set by the support member **30**.
(31) As illustrated in FIGS. **2**, **3**, **4**, and **5**, the support member **30** is provided with a strut portion **31** that extends upward from the vehicle body frame **4** in a state of conforming to a front portion **22***c* of the second exhaust-gas cleaning device **22** and supports the front portion **22***c* of the second exhaust-gas cleaning device **22**, a support arm portion **32** that extends in a vehicle-body forward direction from an upper portion of the strut portion **31** in a state of conforming to a lower portion **21***d* of the first exhaust-gas cleaning device **21** and supports the lower portion **21***d* of the first exhaust-gas cleaning device **21**, and a lower support portion **33** that is provided to a lower portion of the strut portion **31** and supports a lower portion **22***d* of the second exhaust-gas cleaning device **22**.
(32) As illustrated in FIGS. **2**, **4**, and **5**, the strut portion **31** has a washer portion **34** provided to the lower portion of the strut portion **31** and is supported on the vehicle body frame **4** by the washer portion **34** being connected to the vehicle body frame **4**. The connection of the washer portion **34** to the vehicle body frame **4** is performed by the flywheel housing **6** constituting the vehicle body frame **4**. The flywheel housing **6** is positioned below the second exhaust-gas cleaning device **22** and can keep a length of the strut portion **31** short.
(33) As illustrated in FIGS. **2**, **4**, and **5**, the strut portion **31** is provided with left and right strut rods **31***a* lined up at an interval in the vehicle-body horizontal width direction. A lower strut-portion reinforcing rod **31***b*, which connects lower portions of the strut rods **31***a*, and an upper strut-portion

reinforcing rod **31***c*, which connects upper portions of the strut rods **31***a*, are provided across the left and right strut rods **31***a*. The left and right strut rods **31***a* are constituted by steel pipes.
(34) As illustrated in FIGS. **2**, **4**, and **5**, a front support portion **31***d* supporting the second exhaust-gas cleaning device **22** is provided to each of the left and right strut rods **31***a*. The front portion **22***c* of the second exhaust-gas cleaning device **22** is supported by the strut portion **31** by front connecting portions **22***e*—provided in two locations, left and right, in the front portion **22***c* of the second exhaust-gas cleaning device **22**—being connected to the front support portion **31***d*.
(35) As illustrated in FIGS. **2**, **4**, and **5**, washer plates **34***a* provided to respective lower portions of the left and right strut rods **31***a* are provided to the washer portion **34**. In the washer portion **34**, the left and right washer plates **34***a* being connected to the flywheel housing **6** by a plurality of connecting bolts provides a detachable connection to the flywheel housing **6**.
(36) As illustrated in FIGS. **2**, **4**, and **5**, the lower support portion **33** is provided with lower support pieces **33***a* provided to the lower portions of the left and right strut rods **31***a*. The left and right lower support pieces **33***a* are provided to the strut rods **31***a* by being formed on the washer plates **34***a*. The lower support portion **33** is provided to the washer portion **34**. The lower portion **22***d* of the second exhaust-gas cleaning device **22** is supported by the lower support portion **33** by lower connecting portions **22***f*—provided in two locations, left and right, in the lower portion **22***d* of the second exhaust-gas cleaning device **22**— being connected to the lower support pieces **33***a*.
(37) As illustrated in FIGS. **2**, **4**, and **5**, the support arm portion **32** is provided with arm bodies **32***a* that extend in the vehicle-body forward direction from respective upper portions of the left and right strut rods **31***a*. An arm reinforcing rod **32***b* connecting the left and right arm bodies **32***a* is provided across distal end portions of the left and right arm bodies **32***a*. Among the left and right arm bodies **32***a*, the left arm body **32***a* is configured to connect via a relay member **32***c* to a lower connecting portion **21***e* provided to the lower portion **21***d* of the first exhaust-gas cleaning device **21**. The right arm body **32***a* is configured to connect directly to the lower connecting portion **21***e* of the first exhaust-gas cleaning device **21**.
(38) [Configuration of Position Adjustment of Exhaust-Gas Cleaning Devices]
(39) As illustrated in FIGS. **2**, **4**, and **5**, the support member **30** is provided with a position adjustment portion **35** that can adjust a support position—in the vehicle-body vertical direction, the vehicle-body longitudinal direction, and the vehicle-body horizontal width direction—of the second exhaust-gas cleaning device **22** on the strut portion **31** and with a second position adjustment portion **37** that can adjust a support position—in the vehicle-body vertical direction, the vehicle-body longitudinal direction, and the vehicle-body horizontal width direction—of the second exhaust-gas cleaning device **22** on the lower support portion **33**.
(40) When there is a position shift between the exhaust-gas discharging portion **21***b* of the first exhaust-gas cleaning device **21** and the exhaust-gas introduction portion **22***a* of the second exhaust-gas cleaning device **22**, so the positional relationship between the exhaust-gas discharging portion **21***b* and the exhaust-gas introduction portion **22***a* becomes appropriate and the exhaust-gas discharging portion **21***b* and the exhaust-gas introduction portion **22***a* can be appropriately connected by the connecting pipe **23** having no position adjustment function, the positional relationship between the exhaust-gas discharging portion **21***b* and the exhaust-gas introduction portion **22***a* can be adjusted by the position adjustment portion **35** and the second position adjustment portion **37**.
(41) Specifically, as illustrated in FIGS. **2**, **4**, and **5**, the position adjustment portion **35** is provided with an adjustment member **36** positioned between the front connection portion **22***e* of the second exhaust-gas cleaning device **22** and the front support portion **31***d* of the strut portion **31**, a first connecting bolt **36***a* connecting the front connecting portion **22***e* and the adjustment member **36**, and a second connecting bolt **36***b* connecting the front support portion **31***d* and the adjustment member **36**.
(42) As illustrated in FIGS. **2**, **4**, and **5**, the second position adjustment portion **37** is provided with

a second adjustment member **38** positioned between the lower connecting portion **22***f* of the second exhaust-gas cleaning device **22** and the lower support piece **33***a* of the strut portion **31**, a third connecting bolt **38***a* connecting the lower connecting portion **22***f* and the second adjustment member **38**, and a fourth connecting bolt **38***b* connecting the lower support piece **33***a* and the second adjustment member **38**.
(43) A hole diameter of a through hole (not illustrated) of the adjustment member **36** whereinto the first connecting bolt **36***a* is inserted is made greater than an outer diameter of the first connecting bolt **36***a* for a configuration enabling position shifting of the front connecting portion **22***e* relative to the adjustment member **36**. A through hole (not illustrated) of the adjustment member **36** whereinto the second connecting bolt **36***b* is inserted is made greater than an outer diameter of the second connecting bolt **36***b* for a configuration enabling position shifting of the adjustment member **36** relative to the front support portion **31***d*.
(44) A hole diameter of a through hole (not illustrated) of the second adjustment member **38** whereinto the third connecting bolt **38***a* is inserted is made greater than an outer diameter of the third connecting bolt **38***a* for a configuration enabling position shifting of the adjustment member **36** relative to the front support portion **31***d*. A hole diameter of a through hole (not illustrated) of the second adjustment member **38** whereinto the fourth connecting bolt **38***b* is inserted is made greater than an outer diameter of the fourth connecting bolt **38***b* for a configuration enabling position shifting of the adjustment member **36** relative to the front support portion **31***d*.
(45) In the position adjustment portion **35**, among the first connecting bolt **36***a*, the second connecting bolt **36***b*, the third connecting bolt **38***a*, and the fourth connecting bolt **38***b*, tightening of a connecting bolt corresponding to desired position adjustment is loosened. Performing an operation of moving the second exhaust-gas cleaning device **22** shifts the position of the second exhaust-gas cleaning device **22** relative to the support member **30**. This changes the support position—in the vehicle-body vertical direction, the vehicle-body longitudinal direction, and the vehicle-body horizontal width direction—of the second exhaust-gas cleaning device **22** on the strut portion **31** in a manner corresponding to the operation that is performed of moving the second exhaust-gas cleaning device **22**.
(46) In the second position adjustment portion **37**, among the first connecting bolt **36***a*, the second connecting bolt **36***b*, the third connecting bolt **38***a*, and the fourth connecting bolt **38***b*, tightening of a connecting bolt corresponding to desired position adjustment is loosened. Performing an operation of moving the second exhaust-gas cleaning device **22** shifts the position of the second exhaust-gas cleaning device **22** relative to the support member **30**. This changes the support position—in the vehicle-body vertical direction, the vehicle-body longitudinal direction, and the vehicle-body horizontal width direction—of the second exhaust-gas cleaning device **22** on the lower support portion **33** in a manner corresponding to the operation that is performed of moving the second exhaust-gas cleaning device **22**.
(47) [Configuration of Driving Unit]
(48) As illustrated in FIG. **1**, the driving unit **12** is provided behind the engine compartment **15**. As illustrated in FIGS. **1**, **3**, and **8**, the driving unit **12** is provided with the driver's seat **10**, the steering wheel **11** whereby the steering operation of the front wheels **1** is performed, a left brake operation unit **40**L whereby brakes (not illustrated) for the left back wheel are operated, and a right brake operation unit **40**R whereby brakes (not illustrated) for the right back wheel are operated.
(49) [Configurations of Left Brake Operation Unit and Right Brake Operation Unit]
(50) As illustrated in FIG. **8**, the left brake operation unit **40**L and the right brake operation unit **40**R are provided to the right and below the steering wheel **11**. As illustrated in FIGS. **3**, **6**, and **8**, the left brake operation unit **40**L and the right brake operation unit **40**R are each provided with a brake pedal **41**, a return spring **42** that subjects the brake pedal **41** to an operation of being returned to a brake off position (“off”), and a master cylinder **43** connected to the brake pedal **41**.
(51) As illustrated in FIG. **6**, the brake pedal **41** is provided with a pedal arm portion **41***a* and with

a footboard **41***b* provided at a lower end portion of the pedal arm portion **41***a*. A support shaft **44** is provided at an upper portion of the pedal arm portion **41***a*. The support shaft **44** is supported by a support portion **45** provided to the traveling vehicle body **3**. The brake pedal **41** is supported on the traveling vehicle body **3** in a state of being able to swing between the brake off position ("off") and a braking position ("on") by using a shaft core P of the support shaft **44** as a swinging fulcrum. The brake pedal **41** is positioned to the brake off position ("off") by the pedal arm portion **41***a* abutting a stopper **47**.

(52) As illustrated in FIG. **6**, the partition member **17** forming the partition between the engine compartment **15** and the driving unit **12** is disposed further toward a vehicle-body front side than the brake pedal **41**. The return spring **42** is engaged to the pedal arm portion **41***a* and a spring support portion **17***d* provided to the partition member **17**. The return spring **42** biases the brake pedal **41** to swing to the brake off position ("off") by using the partition member **17** as a reaction-force member.

(53) As illustrated in FIG. **6**, the partition member **17** is configured so the part **17***a*, which corresponds to the footboard **41***b*, of the partition member **17** is disposed further toward a vehicle-body front side than the part **17***b*, which corresponds to the support shaft **44**, of the partition member **17**. A front portion of the footwell of the driving unit **12** can be extended forward past the support shaft **44**.

(54) As illustrated in FIG. **6**, the master cylinder **43** is provided on a vehicle-body back side of the support shaft **44** of the brake pedal **41**. The master cylinder **43** is fixed in a detachable manner to the support portion **45**. A slidable operation shaft **43***a* of the master cylinder **43** and an operation arm portion **41***c* provided to the brake pedal **41** are interconnected. The operation arm portion **41***c* extends from the pedal arm portion **41***a* toward an opposite side of a side whereon the footboard **41***b* is positioned relative to the support shaft **44**.

(55) In both the left brake operation unit **40**L and the right brake operation unit **40**R, when the brake pedal **41** is subjected to a stepping operation against the return spring **42** and enters the braking position ("on"), the operation shaft **43***a* of the master cylinder **43** is subjected to a sliding operation to a pushed-in side by the operation arm portion **41***c*. The master cylinder **43** supplies operational hydraulic pressure to the brakes (not illustrated), and the brakes perform an operation of switching to a braking state. When the stepping operation of the brake pedal **41** is released, the brake pedal **41** is subjected to an operation of returning to the brake off position ("off") by the return spring **42**, and the operation shaft **43***a* of the master cylinder **43** is subjected to a sliding operation to a pulled-out side by the operation arm portion **41***c*. The imparting of the operational hydraulic pressure by the master cylinder **43** to the brakes is released, and the brakes perform an operation of switching to an initial state.

(56) [Configuration of Return Spring]

(57) As illustrated in FIGS. **3** and **6**, the return spring **42** is constituted by a torsion coil spring and is provided with a coil portion **42***a*, an arm (a first arm) **42***b* extending from one end side of the coil portion **42***a*, and an arm (a second arm) **42***c* extending from another end side of the coil portion **42***a*. The coil portion **42***a* is fitted onto the support shaft **44**, and the return spring **42** is supported by the support shaft **44**. The arm **42***b* on the one end side is engaged to the spring support portion **17***d* of the partition member **17**. The arm **42***c* on the other end side is engaged to the pedal arm portion **41***a*.

(58) As illustrated in FIG. **6**, a bent end portion **42***d* engaged to the pedal arm portion **41***a* is provided to the arm **42***c* on the other end side. A through hole **46** whereto the bent end portion **42***d* is engaged is provided to the pedal arm portion **41***a*. A shape of the through hole **46** is made to be a shape whereinto the bent end portion **42***d*, which moves along the shaft core P of the support shaft **44**, can be inserted. The return spring **42** and the brake pedal **41** can be assembled by the following assembly outline.

(59) As illustrated in FIG. **7**, the coil portion **42***a* fits the return spring **42** onto the support shaft **44**,

and the arm **42***b* on the one end side is placed in a state of being engaged to the partition member **17**. The return spring **42** is in a free state. The stopper **47** is removed from the support portion **45**. The brake pedal **41** is attached to the support shaft **44** and swung up, and the brake pedal **41** is operated to an attitude of opposing the support shaft, wherein the footboard **41***b* is positioned in a position higher than when the brake pedal **41** is positioned in the brake off position (“off”). The brake pedal **41** in this attitude of opposing the support shaft is guided to the support shaft **44** and moved toward the return spring **42**. By moving the brake pedal **41**, the bent end portion **42***d* of the return spring **42** moves toward the pedal arm portion **41***a* in a direction along the shaft core p of the support shaft **44**, is inserted into the through hole **46** from an inner side of the pedal arm portion **41***a*, and moves to an outer side of the pedal arm portion **41***a*. Next, the brake pedal **41** is swung down. At this time, the operation arm portion **41***c* enters a slit **17***e* provided in the partition member **17**. The stopper **47** is attached to the support portion **45**, and the brake pedal **41** is moved to the brake off position (“off”). This causes the brake pedal **41** to be stopped by the stopper **47**. As illustrated in FIG. **6**, the bent end portion **42***d* engages to the pedal arm portion **41***a*, the return spring **42** is provided with an elastic restoring force, and the brake pedal **41** enters a state of being biased and swung to the brake off position (“off”) by the return spring **42**. The slit **17***e* is closed off once assembly of the brake pedal **41** is finished.

(60) [Configurations of Steering-Post Cover and Panel Cover]

(61) The steering wheel **11** is supported in a state of being able to change positions in the longitudinal direction and the vehicle-body vertical direction. As illustrated in FIG. **8**, a steering-post cover **50** is inserted into a through hole **52** of a panel cover **51** supported on the traveling vehicle body **3**. When the steering wheel **11** changes positions in the longitudinal direction, the steering-post cover **50** follows the steering wheel **11** and swings in the longitudinal direction relative to the panel cover **51**. A bellows-shaped expandable cover **53** is connected across a proximal-side lower portion of the steering-post cover **50** and a lower portion **51***a* of the panel cover **51**. The expandable cover **53** expands and contracts according to the swinging of the steering-post cover **50** but, regardless of the swinging of the steering-post cover **50**, places a gap between the steering-post cover **50** and the lower portion **51***a* of the panel cover **51** in a closed state. Between a part opposing a distal-side wall portion of the steering-post cover **50** of the panel cover **51** and the distal-side wall portion of the steering-post cover **50**, a gap that allows the steering-post cover **50** to swing is provided. However, above the gap and on the distal-side wall portion of the steering-post cover **50**, a bulging portion that bulges toward a cover outer side and makes the gap difficult to see from above is provided. In the panel cover **51**, a face portion **51***b* corresponding to a side portion of the steering-post cover **50** is raised to the proximal side. Even if the steering-post cover **50** swings, the face portion **51***b* makes it difficult to see the gap between the steering-post cover **50** and the panel cover **51** on a horizontal side of the steering-post cover **50**.

Other Embodiments

(62) (1) FIG. **9** is a back view of a panel cover **55** provided with another example of one or more embodiments. FIG. **10** is a perspective view of a steering-post cover **56** provided with another example of one or more embodiments. In the panel cover **55** provided with another example, below a through hole **57** whereinto the steering-post cover **56** is inserted, an opening **58** and a proximal wall **59** extending upward from a lower edge portion of the opening **58** are provided. In the steering-post cover **56**, on a proximal-side lower portion, a proximal extended portion **61** that extends lower than a lower end **60***a* of a horizontal wall portion **60** is provided, and on a distal-side lower portion, a distal extended portion **62** that extends lower than the lower end **60***a* of the horizontal wall portion **60** is provided. The proximal wall **59**, the proximal extended portion **61**, and the distal extended portion **62** serve as hiding walls that make it difficult to see into an area below the panel cover **55** via a gap arising between the steering-post cover **56** and the panel cover **55** on a proximal side and horizontal side of the steering-post cover **56**.

(63) (2) The above embodiments illustrate an example wherein the partition member **17** is

configured so the part **17***a* corresponding to the footboard **41***b* of the brake pedal **41** is disposed further toward the vehicle-body front side than the part **17***b* corresponding to the support shaft **44** of the brake pedal **41**. However, the present invention is not limited thereto. For example, the part **17***a* corresponding to the footboard **41***b* and the part **17***b* corresponding to the support shaft **44** may be positioned in the same position in the vehicle-body longitudinal direction, or the part **17***a* corresponding to the footboard **41***b* may be positioned on a vehicle-body back side of the part **17***b* corresponding to the support shaft **44**.
(64) (3) The above embodiments illustrate an example wherein the return spring **42** is constituted by a coil spring. However, the present invention is not limited thereto, and this may be a tension spring.
(65) (4) The above embodiments illustrate an example of a configuration wherein the bent end portion **52***d* of the return spring **42** can be inserted into the through hole **52** of the pedal arm portion **41***a*. However, the present invention is not limited thereto, and the bent end portion **52***d* may simply be engaged to the through hole **52**.
INDUSTRIAL APPLICABILITY
(66) The present invention can be applied in tractors provided with an engine compartment and with a driving unit that is provided behind the engine compartment and has a brake pedal, a master cylinder operated by the brake pedal, and a return spring that biases the brake pedal to swing back to a brake off position.
(67) Although the disclosure has been described with respect to only a limited number of embodiments, those skilled in the art, having benefit of this disclosure, will appreciate that various other embodiments may be devised without departing from the scope of the present invention. Accordingly, the scope of the invention should be limited only by the attached claims.

Claims

1. A tractor comprising: an engine compartment; a driving unit disposed behind the engine compartment; and a partition that separates the engine compartment and the driving unit from each other, wherein the driving unit comprises: a brake pedal that comprises: a support shaft; and a pedal arm portion; a master cylinder that is operated by the brake pedal and disposed further toward a back side of the tractor than the support shaft; and a return spring that biases the brake pedal to swing back to an initial brake off position; and the partition is disposed further toward a front side of the tractor than the brake pedal, and the return spring is engaged to the partition and to the pedal arm portion.
2. The tractor of claim 1, wherein the brake pedal further comprises: a footboard, and the partition includes a portion that corresponds to the footboard and is disposed further toward the front side than a portion of the partition corresponding to the support shaft.
3. The tractor of claim 1, wherein the return spring is composed of a torsion coil spring comprising a coil portion, and the torsion coil spring is supported by the support shaft with the coil portion engaged with the support shaft.
4. The tractor of claim 3, wherein a first arm extending from one end side of the torsion coil spring is engaged with the partition and a second arm extending from another end side of the torsion coil spring is engaged with the pedal arm portion, the second arm includes: a bent end portion that engages with the pedal arm portion, the pedal arm portion includes: a through hole with which the bent end portion engages, and the bent end portion can be inserted into the through hole in a direction along a shaft core of the support shaft.
5. The tractor of claim 1, wherein the partition includes a vertical portion and an inclined portion inclining toward the engine compartment disposed further toward the front side than the partition.